ModelIt!

Student Activity Pack

Can We Capture Carbon from Thin Air?

Modeling Direct Air Capture Systems, Energy Requirements, and Scalability

Grade 11L3 | NGSS HS-ESS3-4, HS-PS1-4, HS-ETS1-3 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is direct air capture and how is it different from planting trees to absorb CO_2?

2 Why do you think removing CO_2 from the atmosphere is harder than preventing CO_2 emissions in the first place?

3 Draw a diagram showing how you think a machine could pull CO_2 out of regular air and what happens to the CO_2 afterward.

4 If we could build machines to remove CO_2, should we still worry about reducing emissions? Why or why not?

2 Scoring Rubric

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> Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Energy Source Carbon Intensity | Sorbent Efficiency | Thermal Energy Requirement | Net Carbon Removal | Capture Cost | Deployment Scale

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Renewable-Powered DAC

100% renewable energy, current best sorbent technology

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Fossil-Powered DAC

Natural gas powered capture system

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Gigatonne Scale Projection

Scale needed to remove 1 billion tonnes of CO_2 per year

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water Consumption — The amount of water used in the DAC process — some sorbent systems require significant water, creating competition with agricultural and municipal water supplies in arid regions where solar energy is abundant

Sorbent Lifetime — The number of capture-release cycles a sorbent material can undergo before it degrades and must be replaced — longer-lasting sorbents reduce operational costs and waste but may have lower per-cycle efficiency

Storage Permanence — The guaranteed duration of CO_2 storage — geological mineralization is essentially permanent, depleted oil reservoirs last millennia, and enhanced weathering depends on mineral type — permanence determines whether the removal is real or temporary

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If the atmosphere contains too much CO_2, why can't we just build machines to suck it out — and what would it actually take to make a difference? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why the energy source is the most critical variable for whether DAC actually removes carbon from the atmosphere?**

2. **What does the concentration difference between atmospheric CO_2 (420 ppm) and smokestack CO_2 (120,000 ppm) tell you about why DAC is so energy-intensive?**

3. **Why is the moral hazard argument — that DAC reduces urgency for emission reductions — a legitimate policy concern rather than just a philosophical objection?**

4. **How would your model change if a revolutionary sorbent were discovered that captured CO_2 at ten times the efficiency of current materials?**

5. **What are the limitations of modeling DAC deployment without including competing uses for renewable energy like replacing fossil fuel electricity and electrifying transportation?**

8 STEM Challenge: Design a Regional DAC Deployment Strategy

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YOUR ENGINEERING MISSION

Design a 10-year deployment plan for direct air capture in your region, specifying energy source, site selection, storage method, cost reduction pathway, and the tonnes of CO_2 to be removed annually.

THE SCENARIO

Your state government has committed to achieving net-zero emissions by 2040 and has identified direct air capture as one component of the strategy. Your team must design a DAC deployment plan that specifies where to build facilities, what energy source to use, where to store the captured CO_2, how costs will decrease over the 10-year period, and how many tonnes per year the system will remove. The plan must also address whether DAC investment could be better spent on other climate strategies.

Engineering Design Thinking

1. WHAT SPECIFIC DAC TECHNOLOGY WOULD YOU DEPLOY AND WHAT IS ITS CURRENT ENERGY REQUIREMENT PER TONNE OF CO_2?

2. HOW WOULD YOU REDUCE COSTS FROM $600/TONNE TO BELOW $200/TONNE OVER YOUR 10-YEAR DEPLOYMENT PLAN?

3. WHAT IS THE TOTAL RENEWABLE ENERGY CAPACITY NEEDED TO POWER YOUR DAC FACILITIES, AND HOW DOES THIS COMPARE TO YOUR REGION'S CURRENT RENEWABLE CAPACITY?

4. HOW WOULD YOU ADDRESS THE MORAL HAZARD ARGUMENT — THAT DAC INVESTMENT REDUCES URGENCY FOR EMISSION REDUCTIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Carbon Capture Engineers design and optimize direct air capture and industrial carbon capture systems at companies like Climeworks, Carbon Engineering, Global Thermostat, and Heirloom Carbon, earning $90,000-$160,000/year. Climate Policy Analysts who evaluate carbon removal technologies and design government incentive programs earn $80,000-$145,000/year. Chemical Engineers specializing in sorbent development for DAC work in research labs and startups, earning $85,000-$155,000/year.